

Brian Wonjun Lee

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EDUCATION

University of Pennsylvania

MSE in Computer Graphics & Game Technology (CGGT), Submatriculation
BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Philadelphia, PA
Expected May 2028
Expected May 2027

Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

EXPERIENCE

Software Engineer Intern, Aleph Lab (Y Combinator)

June 2026 – Present · San Francisco (Remote)

- Shipped a production retention system that re-engages lapsed users with targeted push notifications (Supabase, Deno edge functions, FCM), measured by a treatment/holdout A/B — drove the product's **first recovered paid subscription**
- Expanded reachable users **~2.7×** by validating the win-back targeting against live production data and correcting how eligible users were identified
- Turned the company's production AI agent into a drop-in package other teams can embed in their own apps — a typed SDK with versioned APIs and **136 automated tests**; a new AI coding agent integrated it into a fresh app on the **first attempt**
- Diagnosed and fixed a production authentication outage (a JWT signing-key migration) that had silently broken the notification pipeline, restoring it **from failing (401) to healthy (200)**

Software Engineer Intern, BitMango (Puzzle1 Studio)

June 2025 – Aug 2025 · Pangyo, South Korea

- Maintained a live portfolio of **50+ Unity mobile game titles**, improving UI/UX and core gameplay stability across releases
- Resolved **100+ QA-reported bugs** by modifying Unity prefabs and C# scripts, shipping fixes in production releases
- Upgraded SDKs and platform modules fleet-wide to meet evolving app-store requirements and sustain production availability

Project Lead & Product Designer, Penn Spark

Jan 2025 – Present · Philadelphia, PA

- Led a cross-functional team of **15+** designers and developers to ship client-facing **0-to-1 products**
- Defined end-to-end user flows, interaction patterns, and reusable interface systems across multiple deliverables
- Shipped polished, user-centered features across semester-long client project cycles

Military Interpreter, Capital Defense Command (ROK Army)

Dec 2022 – June 2024 · Seoul, South Korea

- Provided real-time English–Korean interpretation in high-security, multi-agency environments for international stakeholders
- Translated sensitive documents and briefings with precision, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm

June 2022 – Aug 2022 · Seongnam, South Korea

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG
- Wrote and optimized SQL queries to improve retrieval and management efficiency across large-scale datasets

PROJECTS

Capsule | React, Three.js (R3F), MongoDB, AWS S3

Jan 2025 – Apr 2025

- Built a **full-stack web app** (team of 4) for creating interactive 3D time capsules with media storage and timed unlocking
- Engineered immersive 3D scenes with React Three Fiber for real-time rendering, animation, and direct-manipulation interaction
- Designed the backend on MongoDB with scalable media storage on AWS S3, wiring authentication and timed-release logic

Mini Minecraft | C++, OpenGL, GLSL

Apr 2026

- Engineered a **voxel game engine from scratch** in C++/OpenGL — procedural world generation across **7 biomes** via layered Perlin/FGM/Voronoi noise, plus 3D cave systems and fluid simulation
- Designed **chunk-based streaming with frustum culling** and a data-oriented mesh layout for real-time performance at scale
- Implemented real-time **player physics, voxel collision detection, and interactive block editing** across the generated world

Mini Maya | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- Implemented a **half-edge mesh data structure** enabling efficient traversal and topology-aware editing operations
- Built edge split, extrusion, and **Catmull–Clark subdivision** on an OpenGL/VBO pipeline with GLSL shaders
- Built an interactive **Qt desktop editor** — vertex/edge/face selection, a GUI control panel, and a real-time OpenGL viewport

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, C#, SQL, HTML/CSS, OCaml

Frameworks & Libraries: React, Next.js, Node.js, Astro, Tailwind CSS, Three.js, React Three Fiber, JUnit

Developer Tools: Git, GitLab, AWS, MongoDB, SQLite, Oracle, Figma, VS Code, IntelliJ

Graphics & 3D: OpenGL, GLSL, Unreal Engine 5, Unity, Maya, Blender